2 Answers to Team Problems

Answer 1. 9

Answer 2. 1

Answer 3. $(64, -32, 3)$

Answer 4. $\frac{3}{250}$ (or 0.012)

Answer 5. 6673335

Answer 6. 21

Answer 7. 2062

Answer 8. $\frac{200\sqrt{2923}}{2923}$

Answer 9. 64026

Answer 10. 1029

3 Solutions to Team Problems

Problem 1. Compute the number of ordered triples of positive integers (a, b, c) such that $\sqrt{a^b + c!} = 28$.

Solution 1. Squaring both sides of the given equation gives $a^b + c! = 784$. Accordingly, only values of c up to 6 need be considered, as $7! > 784$.

c	$784 - c!$	prime factorization of $784 - c!$
1	783	$3^3 \cdot 29$
2	782	$2 \cdot 17 \cdot 23$
3	778	$2 \cdot 389$
4	760	$2^3 \cdot 5 \cdot 19$
5	664	$2^3 \cdot 83$
6	64	2^6

For $1 \leq c \leq 5$, the only ordered triple satisfying the equation is $(784 - c!, 1, c)$. For $c = 6$, there are four triples: $(2, 6, 6)$, $(4, 3, 6)$, $(8, 2, 6)$, and $(64, 1, 6)$, for a total of **9** ordered triples.

Problem 2. Chris the frog begins on a number line at 0. Chris takes jumps of lengths $1, 2, 3, \ldots, 2017$, in that order. If Chris's current location is an even integer, he jumps in the positive direction; otherwise, he jumps in the negative direction. Let $P(n)$ denote Chris's location after the nth jump. Compute $\sum_{j=1}^{2017} P(j)$.

Solution 2. List where Chris has landed after his first few jumps.

Jump	Before	Direction	After
1	0	+	1
2	1	−	−1
3	−1	−	−4
4	−4	+	0

In fact, this pattern continues: Chris lands back at the origin after every fourth jump. Suppose that $P(4n) = 0$. Then the next four jumps are as follows.

Jump	Before	Direction	After
$4n+1$	0	+	$4n+1$
$4n+2$	$4n+1$	−	-1
$4n+3$	-1	−	$-4n-4$
$4n+4$	$-4n-4$	+	0

This inductive argument shows that Chris lands back at the origin after every four jumps. Moreover, the sum of the four values of $P(j)$ is constant:

$$\begin{aligned} &P(4n+1) + P(4n+2) + P(4n+3) + P(4n+4) \\ &\quad = (4n+1) + (-1) + (-4n-4) + 0 \\ &\quad = -4. \end{aligned}$$

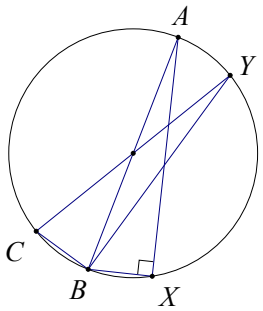
Compute $P(1) + \cdots + P(2017)$ by grouping terms in groups of four:

$$\begin{aligned} \sum_{j=1}^{2017} P(j) &= \left(\sum_{k=0}^{503} \Big(P(4k+1) + P(4k+2) + P(4k+3) + P(4k+4) \Big) \right) + P(2017) \\ &= \left(\sum_{k=0}^{503} -4 \right) + P(2017) \\ &= -2016 + P(2017). \end{aligned}$$

Because 2016 is divisible by 4, after jump 2016, Chris will be at the origin. After jump 2017, Chris will be at 2017, so $P(2017) = 2017$, and the sum is

$$\sum_{j=1}^{2017} P(j) = -2016 + 2017 = \mathbf{1}.$$

Problem 3. The diagram below shows arc $\overset{\frown}{ABC}$, which has a measure of 210°. Points X and Y lie on the arc so that $\text{m}\angle AXB = 90°$ and $\triangle ABX \cong \triangle YCB$. Given that $AX = 8$, the value of $[ABC]$ can be expressed in the form $a + b\sqrt{c}$, where a, b, c are integers and c is not divisible by any perfect square greater than 1. Compute the ordered triple (a, b, c).



Solution 3. Because $\angle AXB$ is a right angle, it follows that $\overline{AB}$ is a diameter of the circle containing points A, B, and C. Thus $\angle ACB$ is also right angle. Also $\angle BAC \cong \angle BYC$ because both angles subtend $\overset{\frown}{BC}$. Because $\angle BYC \cong \angle XAB$, conclude that $\triangle ABC \cong \triangle ABX$. Next, note that $\text{m}\angle XAB = \text{m}\angle BYC = \frac{1}{2} \cdot \text{m}\overset{\frown}{BC} = \frac{1}{2}(210° - 180°) = 15°$. Thus $[ABC] = [ABX] = \frac{1}{2} \cdot AX \cdot BX = \frac{1}{2} \cdot 8 \cdot (8 \tan 15°) = 32 \tan 15°$. The value of $\tan 15°$ can be computed by using the tangent half-angle identity: $\tan(\frac{x}{2}) = \frac{\sin x}{1+\cos x}$ with $x = 30°$, yielding $\tan 15° = \frac{\sin 30°}{1+\cos 30°} = \frac{1/2}{1+\sqrt{3}/2} = 2 - \sqrt{3}$. Thus $[ABC] = 32(2 - \sqrt{3}) = 64 - 32\sqrt{3}$, and the desired ordered triple (a, b, c) is **(64, −32, 3)**.

Note: The value of $\tan 15°$ can also be computed using the identity $\tan(A - B) = \frac{\tan A - \tan B}{1+\tan A \tan B}$, with $A = 45°$ and $B = 30°$.

Alternate Solution: Proceed as in the previous solution to conclude that $\triangle ABC \cong \triangle ABX$. Then $AB = \frac{AX}{\cos 15°}$ and $[ABC] = \frac{1}{2} \cdot AX \cdot \frac{AX}{\cos 15°} \cdot \sin 15°$. Use the subtraction identities to compute $\cos 15° = \cos(45° - 30°) = \cos 45° \cos 30° + \sin 45° \sin 30° = \frac{\sqrt{6}+\sqrt{2}}{4}$ and $\sin 15° = \sin(45° - 30°) = \sin 45° \cos 30° - \cos 45° \sin 30° = \frac{\sqrt{6}-\sqrt{2}}{4}$. Dividing, $\frac{\sin 15°}{\cos 15°} = \frac{\sqrt{6}-\sqrt{2}}{\sqrt{6}+\sqrt{2}} = \frac{(\sqrt{6}-\sqrt{2})(\sqrt{6}-\sqrt{2})}{(\sqrt{6}+\sqrt{2})(\sqrt{6}-\sqrt{2})} = \frac{6-2\sqrt{12}+2}{4} = 2 - \sqrt{3}$. Substituting into the expression for $[ABC]$ yields $\frac{1}{2} \cdot 8^2 \cdot \left(2 - \sqrt{3}\right) = 64 - 32\sqrt{3}$, as in the previous solution.

Problem 4. Ari repeatedly rolls a standard, fair, six-sided die. Let $R(n)$ be the nth number rolled, and let $Q(n) = R(1) \cdot R(2) \cdot \ldots \cdot R(n)$. Compute the probability that there exists an n such that $Q(n) = 100$ and for all $m < n$, $Q(m)$ is not a perfect square.

Solution 4. In order for the conditions in the problem to be met, the first roll must be either 2 or 5. After the first roll, rolls of 1 can be ignored, because they are essentially equivalent to simply re-rolling; there are five other possibilities. Because of the requirement that no partial product before 100 be a perfect square, there are only five sequences of non-1 rolls that result in success: 2-5-2-5, 2-5-5-2, 5-4-5, 5-2-5-2, and 5-2-2-5. The probability of obtaining each sequence of length four is $\frac{1}{6} \cdot \left(\frac{1}{5}\right)^3 = \frac{1}{750}$, while the probability of obtaining the

sequence of length three is $\frac{1}{6} \cdot \left(\frac{1}{5}\right)^2 = \frac{5}{750}$. Because there are four sequences of length four and one sequence of length three, the desired probability is $4 \cdot \frac{1}{750} + 1 \cdot \frac{5}{750} = \frac{9}{750} = \boxed{\frac{3}{250}}$.

Problem 5. Given that C, A, T, F, I, S, and H are digits, not necessarily distinct, and that

$$3 \cdot \underline{C}\ \underline{A}\ \underline{T} \cdot \underline{F}\ \underline{I}\ \underline{S}\ \underline{H} = \underline{C}\ \underline{A}\ \underline{T}\ \underline{F}\ \underline{I}\ \underline{S}\ \underline{H},$$

compute the greatest possible value of $\underline{C}\ \underline{A}\ \underline{T}\ \underline{F}\ \underline{I}\ \underline{S}\ \underline{H}$.

Solution 5. First rewrite the given equation as

$$3 \cdot \underline{C}\ \underline{A}\ \underline{T} \cdot \underline{F}\ \underline{I}\ \underline{S}\ \underline{H} = 10000 \cdot \underline{C}\ \underline{A}\ \underline{T} + \underline{F}\ \underline{I}\ \underline{S}\ \underline{H}$$

which gives

$$3 = \frac{10000}{\underline{F}\ \underline{I}\ \underline{S}\ \underline{H}} + \frac{1}{\underline{C}\ \underline{A}\ \underline{T}}.$$

Therefore, in order to compute the greatest possible value of $\underline{C}\ \underline{A}\ \underline{T}$, and hence of $\underline{C}\ \underline{A}\ \underline{T}\ \underline{F}\ \underline{I}\ \underline{S}\ \underline{H}$, $\frac{10000}{\underline{F}\ \underline{I}\ \underline{S}\ \underline{H}}$ must be as close as possible to 3 without going over, which means $\underline{F}\ \underline{I}\ \underline{S}\ \underline{H}$ must be as close to 3333 as possible. Trying $\underline{F}\ \underline{I}\ \underline{S}\ \underline{H} = 3334$ gives $\underline{C}\ \underline{A}\ \underline{T} = 1667$, which is impossible. Trying $\underline{F}\ \underline{I}\ \underline{S}\ \underline{H} = 3335$ gives $\underline{C}\ \underline{A}\ \underline{T} = 667$, which is the greatest possible value of $\underline{C}\ \underline{A}\ \underline{T}$.

Thus the greatest possible value for $\underline{C}\ \underline{A}\ \underline{T}\ \underline{F}\ \underline{I}\ \underline{S}\ \underline{H}$ is **6673335**.

Problem 6. Let $\{a_n\}$ be a sequence with $a_0 = 1$, and for all $n > 0$, $a_n = \frac{1}{2}\sum_{i=0}^{n-1} a_i$. Compute the greatest value of n for which $a_n < 2017$.

Solution 6. Examine the first few terms in the sequence until a pattern emerges.

$$\begin{aligned} a_1 &= \frac{1}{2} = \frac{3^0}{2^1} \\ a_2 &= \frac{1}{2}\left(1 + \frac{1}{2}\right) = \frac{3}{4} = \frac{3^1}{2^2} \\ a_3 &= \frac{1}{2}\left(1 + \frac{1}{2} + \frac{3}{4}\right) = \frac{9}{8} = \frac{3^2}{2^3} \end{aligned}$$

The following argument shows that for $n \geq 1$, $a_n = \frac{3^{n-1}}{2^n}$. The base case is shown above. For the induction hypothesis, assume that $a_k = \frac{3^{k-1}}{2^k}$ for some $k \geq 1$; then use the geometric series sum formula with $r = \frac{3}{2}$ to compute a_{k+1}:

$$\begin{aligned} a_{k+1} &= \frac{1}{2}(a_0 + a_1 + \cdots + a_k) \\ &= \frac{1}{2}\left(1 + \frac{3^0}{2^1} + \frac{3^1}{2^2} + \frac{3^2}{2^3} + \cdots + \frac{3^{k-1}}{2^k}\right) \\ &= \frac{1}{2}\left(1 + \frac{1}{2}\left(\frac{1 - \frac{3^k}{2^k}}{1 - \frac{3}{2}}\right)\right) = \frac{1}{2}\left(1 - \frac{1}{2}\left(\frac{1 - \frac{3^k}{2^k}}{\frac{1}{2}}\right)\right) = \frac{1}{2}\left(\frac{3^k}{2^k}\right) = \frac{3^k}{2^{k+1}}. \end{aligned}$$

Thus for $n > 0$, $a_n = \frac{3^{n-1}}{2^n} = \frac{1}{2} \cdot \left(\frac{3}{2}\right)^{n-1}$.

Consequently, the number which must be computed is the greatest value of n for which $1.5^{n-1} < 2 \cdot 2017 = 4034$, i.e., $1.5^n < 6051$. Note that $1.5^4 = 2.25^2 = 5.0625 > 5$. Therefore $1.5^{20} = (1.5^4)^5 > 5^5 = 3125$, and so

$1.5^{22} > 3125 \cdot 2.25$ which is clearly larger than 6051. Thus the desired value of n is no more than 21. Moreover, because 1.5^4 is only slightly larger than 5 and $1.5^{20} = \left(1.5^4\right)^5$, one should expect that 1.5^{20} will be only slightly larger than $5^5 = 3125$. This argument suggests that $3125 \cdot 1.5 = 4627.5$ is a reasonable approximation for 1.5^{21}, making 21 a reasonable guess for the greatest n with $1.5^n < 6051$.

To verify this guess, it suffices to verify that $1.5^{20} < 4034$. Expanding 1.5^{20} using the binomial theorem gives

$$\begin{aligned} 1.5^{20} &= (1.5^4)^5 \\ &= (5 + 0.0625)^5 \\ &= \binom{5}{0} \cdot 5^5 + \binom{5}{1} \cdot 5^4 \cdot 0.0625 + \binom{5}{2} \cdot 5^3 \cdot 0.0625^2 + \\ &\quad \binom{5}{3} \cdot 5^2 \cdot 0.0625^3 + \binom{5}{4} \cdot 5 \cdot 0.0625^4 + \binom{5}{5} \cdot 0.0625^5. \end{aligned}$$

The largest binomial coefficient is $\binom{5}{2} = \binom{5}{3} = 10$, but $0.0625 < 0.1$, so when positive powers of 0.0625 are multiplied by binomial coefficients, the result is never more than 1. Replacing the binomial coefficients and powers of 0.0625 with 1 yields an upper bound of

$$1.5^{20} < 5^5 + 5^4 + 5^3 + 5^2 + 5^1 + 5^0 = \frac{5^6 - 1}{5 - 1} = 3906 < 4034.$$

Thus $1.5^{21} < 4034 \cdot 1.5 = 6051$, and so **21** is indeed the correct answer.

Problem 7. On July 17, 2017, the nation of Armlandia will turn n^2 years old and the nation of Nysmlistan will turn n years old. The next four anniversaries for which Nysmlistan's age divides Armlandia's age will occur, in order, on July 17 in the years 2027, 2032, 2038, and M. Compute M.

Solution 7. First note that the age of Nysmlistan in each of 2027, 2032, and 2038 is $n + 10$, $n + 15$, and $n + 21$, respectively. Similarly, the age of Armlandia in those years is $n^2 + 10$, $n^2 + 15$, and $n^2 + 21$. Next suppose that $n + k \mid n^2 + k$, for some integers n and k. Then using polynomial long division,

$$n^2 + k = (n - k)(n + k) + \frac{k(k + 1)}{n + k}$$

which implies that $n + k$ must divide $k(k + 1)$.

Therefore $n + 10$ must divide 110, $n + 15$ must divide 240, and $n + 21$ must divide 462. Factoring each value shows that only two values of n satisfy these three conditions: $n = 1$ and $n = 45$. The condition that 2027 is the first year after 2017 in which Nysmlistan's age divides Armlandia's age precludes $n = 1$, so $n = 45$.

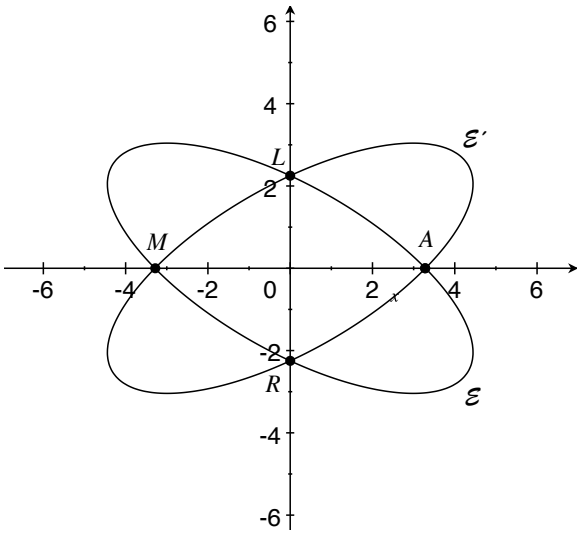
Let $m = M - 2017$, so that Nysmlistan's age in year M is $m + n$. Because $M > 2038$, $m + n > 21$. Then, following the above, $n + m \mid (n^2 + m)$, so $n + m$ divides $m(m + 1)$. Substituting $n = 45$ gives $m + 45 \mid (m^2 + m)$, and using polynomial long division once again yields

$$m^2 + m = m - 44 + \frac{1980}{m + 45}.$$

The first factors of 1980 greater than 45 are 55, 60, 66, and 90, corresponding to $m = 10$, $m = 15$, $m = 21$, and $m = 45$, respectively. Hence $m = 45$ and $M = \mathbf{2062}$.

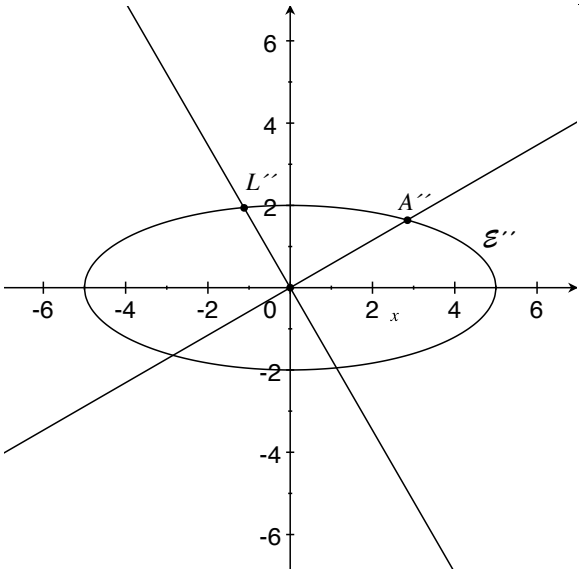
Problem 8. Ellipse $\mathcal{E}$ has center O, major axis of length 10, and minor axis of length 4. Ellipse $\mathcal{E}'$ is obtained by rotating $\mathcal{E}$ counterclockwise about O by 60°. Proceeding clockwise around the perimeter of $\mathcal{E}$, the intersection points of $\mathcal{E}$ and $\mathcal{E}'$ are labeled A, R, M, L. Compute $[AOL]$.

Solution 8. Place the ellipse and its images on the coordinate plane. While ordinarily it is more convenient to place ellipses so that their major and minor axes are parallel to (or coincide with) the x- and y-axes, in this situation, it's more strategic to place the ellipses so that their *intersection points* lie on the axes. Accordingly, place $\mathcal{E}$ on the xy-plane so that its center is at the origin and its major axis is rotated 30° clockwise relative to the x-axis. Then $\mathcal{E}'$ will also have its center at the origin, but will have its major axis rotated 30° counterclockwise relative to the x-axis. Because this diagram is symmetric about both $x-$ and $y-$ axes, the points of intersection A, R, M, and L all lie on the coordinate axes, as shown in the diagram below.



Without loss of generality, let $A = (a, 0)$ be the point of intersection on the positive x-axis. Then $L = (0, \ell)$ is the point of intersection on the positive y-axis, and $[AOL] = \frac{a\ell}{2}$.

To compute the lengths a and ℓ, rotate $\mathcal{E}$ by 30° counterclockwise to a new ellipse $\mathcal{E}''$ so that its major axis coincides with the x-axis. The points A and L will also be rotated by 30° to new points A'' and L'', respectively. Because $\triangle AOL \cong \triangle A''OL''$, it follows that $[AOL] = [A''OL''] = \frac{1}{2}A''O \cdot L''O$. Now A'' is the intersection of $\mathcal{E}''$ and a line through the origin making a 30° angle with the x-axis, as shown in the diagram below.



The equation for $\mathcal{E}''$ is $\dfrac{x^2}{25} + \dfrac{y^2}{4} = 1$, or $4x^2 + 25y^2 = 100$. Because $\tan 30^\circ = \frac{1}{\sqrt{3}}$, the equation for the line is

$y = \frac{1}{\sqrt{3}}x$. Substitute to obtain

$$\begin{aligned} 4x^2 + 25\left(\frac{1}{\sqrt{3}}x\right)^2 &= 100 \\ 12x^2 + 25x^2 &= 300 \\ x^2 &= \frac{300}{37} \\ x &= 10\sqrt{\frac{3}{37}}. \end{aligned}$$

Because $a = A''O$, it suffices to compute $A''O$ using trigonometry: $\frac{x}{A''O} = \cos 30^\circ = \frac{\sqrt{3}}{2}$, so $A''O = \frac{10\sqrt{\frac{3}{37}}}{\frac{\sqrt{3}}{2}} = \frac{20}{\sqrt{37}}$.

Similarly, L'' is the intersection of $\mathcal{E}''$ with the line $y = -\sqrt{3}x$. Solve for the intersection point to obtain

$$\begin{aligned} 4x^2 + 25\left(-\sqrt{3}x\right)^2 &= 100 \\ 79x^2 &= 100 \\ x &= -\frac{10}{\sqrt{79}}. \end{aligned}$$

Because $\ell = L''O$ and $\frac{x}{L''O} = \cos 120^\circ = -\frac{1}{2}$, conclude that $\ell = \frac{20}{\sqrt{79}}$.

Then $[AOL] = [A''OL''] = \dfrac{a\ell}{2} = \dfrac{\left(\frac{20}{\sqrt{37}}\right)\left(\frac{20}{\sqrt{79}}\right)}{2} = \boxed{\dfrac{200\sqrt{2923}}{2923}}$.

Problem 9. Let $E(n)$ denote the least integer strictly greater than n whose base-10 representation contains only even digits, and let $O(n)$ denote the least integer strictly greater than n whose base-10 representation contains only odd digits. Compute the least positive integer N for which $E(N) + O(N)$ is a multiple of 2017.

Solution 9. Let $S(n) = E(n) + O(n)$. In general, either $S(n)$ consists of only odd digits, or $S(n)$ has initial digit 2, and then only odd digits. To prove this observation, notice that one of $E(n)$ and $O(n)$ will consist of an initial digit, and then either all 0's or all 1's, respectively: if d is the first (leftmost) digit of n and d is odd, then the next larger number with all even digits must have first digit $d + 1$ (or 2 if $d = 9$) followed by all 0's, while if d is even, then the next larger number with all odd digits must have first digit $d + 1$ followed by all 1's. To compute the sum of $E(n)$ and $O(n)$, consider the sum of each pair of corresponding digits separately, beginning with the digits immediately to the right of their leftmost digits. If d is odd, this process is equivalent to adding 0's to $O(n)$'s digits. If d is even, this process is equivalent to adding 1's to $E(n)$'s digits, and none of $E(n)$'s digits is a 9. In neither case is there any need to carry. Thus when the $E(n)$ and $O(n)$ are added, there will be no carrying, except possibly in the initial digit. If $E(n)$ and $O(n)$ have the same number of digits, then the initial digit of $S(n)$ will be odd. If $E(n)$ and $O(n)$ do not have the same number of digits, then $O(n)$ must have initial digit 9 and $E(n)$ will have one more digit than $O(n)$, with an initial 2 and 0's everywhere else. Either way, $S(n)$ will fit the above form. This argument proves the lemma.

Therefore, to determine possible values of $S(N)$, look for multiples of 2017 that either consist of only odd digits or have an initial 2 and then all odd digits. Consider the thousands place of the first few multiples of 2017, which are $2017, 4034, 6051, \ldots$. The thousands place will continue to be even, until enough multiples of 17 are accumulated to roll over into the thousands place. As long as the thousands place is even, the multiple of 2017 cannot be a possible value of $S(N)$, because all the digits will not be even, and the thousands place is not the initial digit of the multiple (except for 2017, which does not work because 0 is not odd). Therefore, noting that $58 < 1000/17 < 59$, skip directly to $59 \cdot 2017 = 119003$. In order to force the hundreds place to be

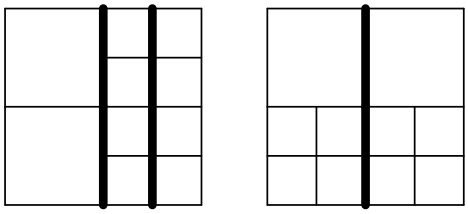
odd, note that $5 < 97/17 < 6$, so skip ahead by $6 \cdot 2017$ to $65 \cdot 2017 = 131105$. This value still doesn't work, so noting that $S(N)$ must be odd, consider $S(N) = 67 \cdot 2017 = 135139$.

Note that $S(N)$ does not begin with a 2, so $E(N)$ and $O(N)$ have the same number of digits, and the first digits of $E(N)$ and $O(N)$ must differ by 1, so $E(N)$ must begin with a 6 and $O(N)$ must equal 71111. Thus $E(N) = 135139 - 71111 = 64028$, and hence $N = \min(64028, 71111) - 2 = \mathbf{64026}$.

Problem 10. Compute the number of tilings of a 4×7 rectangle using only 1×1, 2×2, 3×3, and 4×4 tiles.

Solution 10. Let $T(n)$ be the number of ways to tile a $4 \times n$ rectangle with the given types of tiles, with $T(0) = 1$. Because there is only one way to tile a 4×1 rectangle, i.e., with 1×1 squares, $T(1) = 1$.

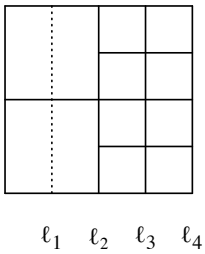
Use the term *fault line* to describe a vertical line from top to bottom that does not intersect any tiles and that is not either the left or the right border of the figure. For example, if a 4×4 region is tiled with two 2×2 tiles on the left, and then eight 1×1 tiles, as shown in the left figure below, then the tiling has two fault lines, while if the two 2×2 tiles are next to each other at the top of the region, then the tiling has only one fault line, as shown in the right figure below.



Two fault lines One fault line

Under this definition, it is possible for a tiling to have no fault lines; for example, if the region is completely filled by a single square, or if 2×2 squares are staggered so that the right half of one is directly above or below the left half of the other. Thus, in general, a $4 \times n$ rectangle can have between 0 and $n-1$ fault lines, inclusive. The number of tilings can be counted by conditioning on the location of the leftmost fault line.

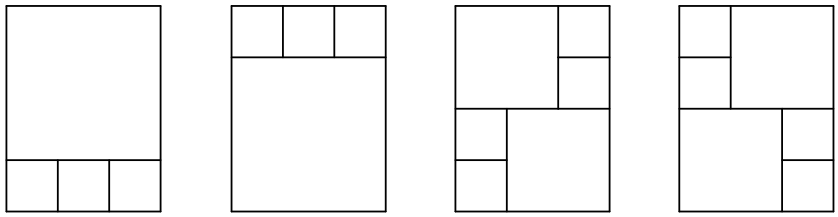
For ease of notation, label the vertical lines (whether or not they are fault lines) one unit to the right of the leftmost edge, two units to the right of the leftmost edge, etc. as ℓ_1, ℓ_2, etc. respectively, as shown below. In that example, the leftmost fault line is at ℓ_2.



Let $F(n)$ represent the number of ways to tile a $4 \times n$ rectangle with *no* fault lines. Note that $F(0) = T(0) = 1$ and $F(1) = T(1) = 1$.

A 4×2 rectangle can be tiled with no fault line in four ways (two 2×2 squares or one 2×2 square and four 1×1 squares), so $F(2) = 4$. If there is one fault line, it is at ℓ_1, and divides the rectangle into two 4×1 strips, each of which can be tiled in $F(1) = T(1)$ ways, for $F(1) \cdot T(1) = 1$ tiling. So $T(2) = F(2) + F(1) \cdot T(1) = 5$.

A 4×3 rectangle can be tiled with no fault line in four ways: either there is a 3×3 square (and the rest of the area is filled in with 1×1 squares), which can happen in two ways, or there are two 2×2 squares, one above and to either the left or right of the other, with the other spaces filled in with 1×1 squares.



Hence $F(3) = 4$. If the leftmost fault line is at ℓ_1, then there are $F(1)$ ways to tile the left side and $T(2)$ ways to tile the right side. If the leftmost fault line is at ℓ_2, then there are $F(2)$ ways to tile the left side and $T(1)$ ways to tile the right side. Hence $T(3) = F(3) + F(1) \cdot T(2) + F(2) \cdot T(1) = 4 + 1 \cdot 5 + 4 \cdot 1 = 13$.

A 4×4 rectangle can be tiled with no fault line in three ways: either using a single 4×4 square, or by alternating 2×2 squares in the top and bottom row (and filling in with four 1×1 squares on the ends) as in the 4×3 case, which can happen in two ways. So $F(4) = 3$. Otherwise, with the first fault line occurring at ℓ_1, ℓ_2, or ℓ_3, there are $F(1) \cdot T(3)$, $F(2) \cdot T(2)$, and $F(3) \cdot T(1)$ possible tilings, respectively, so that

$$\begin{aligned} T(4) &= F(4) + F(1) \cdot T(3) + F(2) \cdot T(2) + F(3) \cdot T(1) \\ &= 3 + 1 \cdot 13 + 4 \cdot 5 + 4 \cdot 1 \\ \implies T(4) &= 40. \end{aligned}$$

The foregoing analysis shows that, in general,

$$T(n) = F(n) + F(1) \cdot T(n-1) + F(2) \cdot T(n-2) + \cdots + F(n-1) \cdot T(1).$$

Indeed, the proof is trivial, because if there is no fault line, there are $F(n)$ tilings, and if the leftmost fault line is at ℓ_i, then there are $F(i)$ ways of tiling the region to the left of the fault line without any fault lines, and there are $T(n-i)$ ways of tiling the region to the right of the fault line (with or without fault lines). Also, for $n > 4$, $F(n) = 2$, because the only way to tile such a region without fault lines is to alternate 2×2 squares in the top and bottom rows. So a simpler version of the formula above for $n > 4$ is simply

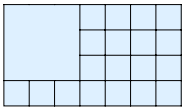
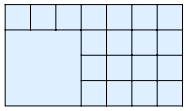
$$T(n) = 1 \cdot T(n-1) + 4 \cdot T(n-2) + 4 \cdot T(n-3) + 3 \cdot T(n-4) + 2 \cdot T(n-5) + \cdots + 2 \cdot T(0).$$

Using this formula, construct a table of values, as shown below.

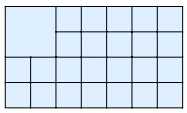
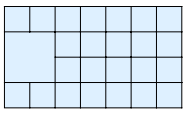
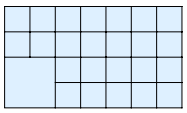
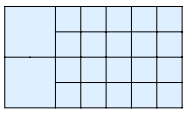
n	0	1	2	3	4	5	6	7
$T(n)$	1	1	5	13	40	117	348	**1029**

Alternate Solution: Let $T(n)$ be the number of ways to tile an $4 \times n$ rectangle with square integer-length tiles, and let $S(n,k)$ be the number of ways to tile the same rectangle conditional on the largest tile touching the left-hand border being of size $k \times k$. Then $T(n) = S(n,1) + S(n,2) + S(n,3) + S(n,4)$.

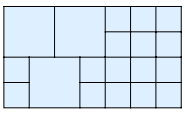
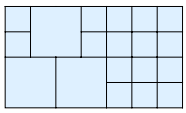
Because there is only one way to arrange a strip of 1×1 squares along the left side, $S(n,1) = T(n-1)$. Because there is only one way to arrange a single 4×4 square on the left side, $S(n,4) = T(n-4)$. As the diagram below shows, there are two ways to arrange a 3×3 square and three 1×1 squares so that the 3×3 square touches the left side, so $S(n,3) = 2 \cdot T(n-3)$.

The remaining term to address is the $S(n, 2)$ term. There are three ways to fit one 2×2 tile and four 1×1 tiles in the leftmost 4×2 subregion. There is also one way to fit two 2×2 tiles in that subregion, as illustrated below.

However, it is also possible for a series of two or more 2×2 squares to alternate across an $4 \times m$ range, where m can be any integer larger than 2. Note that for any such m, this alternating set of 2×2 squares can occur in two ways, depending on whether the leftmost 2×2 square is at the top or the bottom. The diagram below shows both examples with $m = 4$.

Thus $S(n, 2) = 4 \cdot T(n-2) + 2 \cdot \sum_{m \geq 3} T(n-m)$.

Combine the above results to obtain the following recursion:

$$T(n) = T(n-1) + 4 \cdot T(n-2) + 2 \cdot T(n-3) + T(n-4) + 2 \cdot \sum_{m \geq 3} T(n-m).$$

Use the recursion to build the table below.

n	0	1	2	3	4	5	6	7
$T(n)$	1	1	5	13	40	117	348	1029

Hence there are **1029** tilings. They are illustrated on the following page.

